

```
usman@Shaukat: ~/xv6
File Edit View Aa ...
usman Shaukat 31555
Ln 1, Col 20 | 19 character Plain t | 100% | Wir UTF-8

/_init user/_kill
ts user/_grind use
nmeta 47 (boot, su
balloc: first 1127
balloc: write bitm
qemu-system-riscv6
egacy=false -drive

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ sleep 10
$ sleep
usage: sleep ticks
$

/_rm user/_sh user/_sixfive user/_sleep
user/_forphan user/_dorphan user/_sync
3, bitmap blocks 1) blocks 1953 total 20

kernel/kernel -m 128M -smp 3 -nographic
=x0 -device virtio-blk-device,drive=x0,b
```

```
$ memdump
Example 1:
61810
2026
Example 2:
a string
Example 3:
another
Example 4:
C50
1819438967
100
z
xyzzzy
Example 5:
hello
w
o
r
l
d
$

usman Shaukat 31555
Ln 1, Col 20 | 19 character Plain t | 100% | Wir UTF-8
```

```
$ sixfive sixfive.txt
5
100
18
6
$

$ echo > b
mkdir a
echo > a/b
mkdir a/aa
echo > a/aa/b$ $ $ $ find . b
$ find . b
./b
./a/b
$ find . wc -exec echo hi
hi ./wc
$

Usman [ERP: 31555] ~/xv6-labs-2026$ ./grade-lab-util find
make: 'kernel/kernel' is up to date.
== Test find, in current directory == find, in current directory: OK (2.8s)
== Test find, in sub-directory == find, in sub-directory: OK (1.7s)
== Test find, recursive == find, recursive: OK (2.0s)
== Test find, missing target, no duplicates, directory match == find, missing target, no duplicates, directory match: OK (2.1s)
Usman [ERP: 31555] ~/xv6-labs-2026$ ./grade-lab-util sleep
make: 'kernel/kernel' is up to date.
== Test sleep, no arguments == sleep, no arguments: OK (1.8s)
== Test sleep, returns == sleep, returns: OK (1.3s)
== Test sleep, makes syscall == sleep, makes syscall: OK (1.2s)
== Test sleep, zero and non-numeric args return == sleep, zero and non-numeric args return: OK (1.2s)
Usman [ERP: 31555] ~/xv6-labs-2026$ ./grade-lab-util sixfive
make: 'kernel/kernel' is up to date.
== Test sixfive_test == sixfive_test: OK (2.1s)
== Test sixfive_readme == sixfive_readme: OK (1.7s)
== Test sixfive_all == sixfive_all: OK (1.7s)
== Test sixfive, stdin and missing file == sixfive, stdin and missing file: OK (1.3s)
Usman [ERP: 31555] ~/xv6-labs-2026$ ./grade-lab-util memdump
make: 'kernel/kernel' is up to date.
== Test memdump, examples == memdump, examples: OK (2.2s)
== Test memdump, format ii, S, p == memdump, format ii, S, p: OK (1.3s)
== Test memdump, formats h c, bounds checking, usage == memdump, formats h c, bounds checking, usage: OK (1.3s)
Usman [ERP: 31555] ~/xv6-labs-2026$
```

optional exercises:

```
balloc: first 1166 blocks have been allocated
balloc: write bitmap block
qemu-system-riscv64 -machine ... kernel -m 12
drive file=fs.img,if=none, ... -device,dri

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ uptime
1836
$ uptime
1883
$
```

```
usman Shaukat 31555

took a break isliye itna uptime aaraha
lol

Ln 4, Col 43 | 64 character | Plain t | 100% | Wir | UTF-8
```

```
$ uptime
1836
$ uptime
1883
$ echo > b
$ mkdir a
$ echo > a/b
$ mkdir a/aa
$ echo > a/aa/b
$ find .^s
usage: find path name [-exec cmd args...]
$ find . ^s
./sixfive.txt
./sh
./sixfive
./sleep
./stressfs
./sync
$ find . c$
./wc
./sync
$ find . .c
./echo
./wc
./sync
$ find . s.*p
./sleep
$ find . b
./zombie
./b
./a/b
./a/aa/b
$ find . wc -exec echo hi
```

```
usman Shaukat 31555

Ln 4, Col 1 | 22 character | Plain t | 100% | Wir | UTF-8
```